



	EE 354/CE 361/MATH 310 - Introduction to Probability and Statistics Fall 2025				
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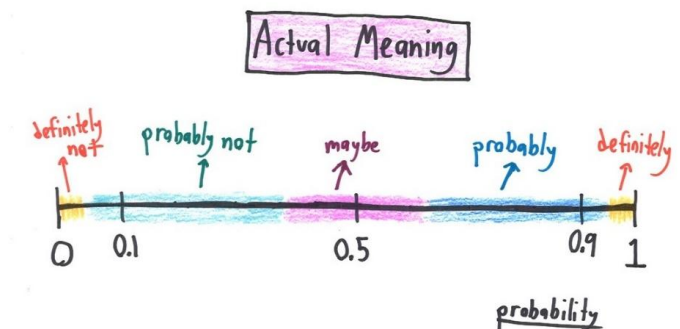


EE 354/CE 361/MATH 310 L2 section (Fall 2025)

Introduction to Probability and Statistics -

Aamir Hasan

Week 5 - Lecture No 08 – 17th September 2025



Announcements!

1. Lectures up to No 07 posted
2. Additional lecture - September 20, 2025 (Saturday, same time & place) in-lieu of the class missed on Wednesday August 20, 2025
3. An added lecture will be posted in a asynchronous mode in lieu of September 08, 2025 class
4. **Quiz No 02 – September 17, 2025 (Conditional Probability and Bayes Rule)**
5. **Quiz No 03 – September 17, 2025 (Counting)**

Agenda for today

1. Unit 3 - Counting

Example

- Find the probability that:
six rolls of a (six sided) die all give different numbers.
(Assume all outcomes equally likely.)

A

→ Typical outcome: $P(2, 3, 4, 5, 6, 2) = \frac{1}{6^6}$

→ Typical element of A: $(2, 3, 4, 1, 6, 5) = 6!$

$$\rightarrow P(A) = \frac{\# \text{ in } A}{\# \text{ possible outcomes}} = \frac{6!}{6^6}$$

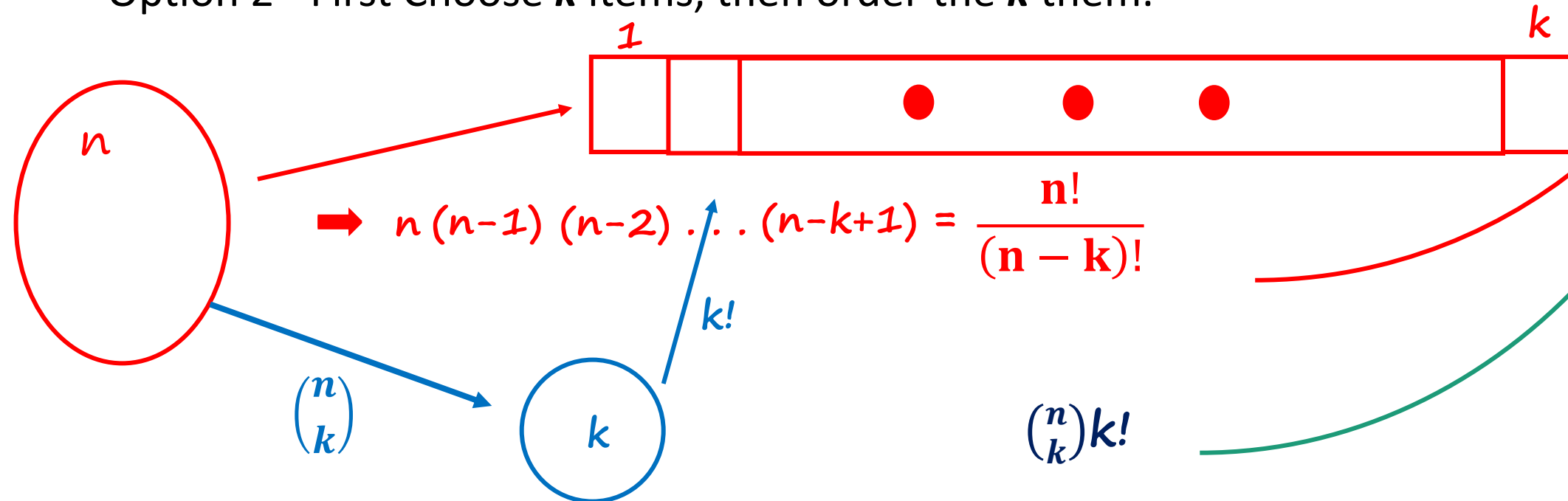
Combinations

Definition: $\binom{n}{k}$ Number of k -element subsets of a given n -element set
 $\underline{\underline{=}} \quad \binom{n}{k} = \frac{n!}{k!(n-k)!}$

→ $n = 0, 1, 2, \dots$

→ $k = 0, 1, 2, \dots, m$

- Two ways of constructing an **ordered** sequence of k **distinct** items:
 - Option 1 - Choose the k items one at a time.
 - Option 2 - First Choose k items, then order the k them.



$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

$$\rightarrow \binom{n}{n} = 1 \quad \frac{n!}{n! 0!} \quad 0! = 1 \text{ convention}$$

$$\rightarrow \binom{n}{0} = \frac{n!}{0! n!} = 1 \quad \emptyset$$

$$\rightarrow \sum_{k=0}^n \binom{n}{k} = \binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{n} = \# \text{ all subsets} = 2^n$$

Binomial coefficient $\binom{n}{k} \rightarrow$ Binomial probabilities

- $n \geq 1$ independent coin tosses;

$$P(H) = p$$

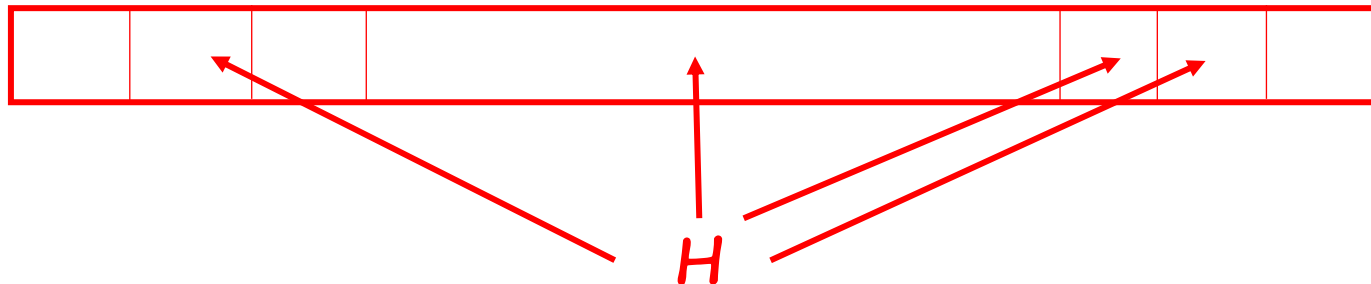
$$P(k \text{ heads}) = \binom{n}{k} p^k (1-p)^{n-k}$$

- $P(H T T H H H) = P(1-p)(1-p)p p p = p^4 (1-p)^2$

- $P(\text{particular sequence}) = p^{\# \text{heads}} (1-p)^{\# \text{tails}}$

- $P(\text{particular } k\text{-head sequence}) = p^k (1-p)^{n-k}$

$$\Rightarrow P(k \text{ heads}) = p^k (1-p)^{n-k} * (\# \text{ of } k\text{-head sequence})$$



$$\binom{n}{k}$$

A coin tossing problem

- Given that there were 3 heads in 10 tosses, what is the probability that the first two tosses were heads?

Assumptions:

- independence
- $P(H) = p$

$$P(k \text{ heads}) = \binom{n}{k} p^k (1-p)^{n-k}$$

- event A : the first 2 tosses were heads.
- event B : 3 out of 10 tosses were heads.
- First solution:

$$\rightarrow P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{P(H_1 H_2 \text{ and one } H \text{ in tosses } 3, \dots, 10)}{P(B)}$$

$$\rightarrow = \frac{p^2 \cdot \binom{8}{1} p^1 \cdot (1-p)^7}{\binom{10}{3} p^3 (1-p)^7} = \frac{\binom{8}{1}}{\binom{10}{3}} = \frac{8}{\binom{10}{3}}$$

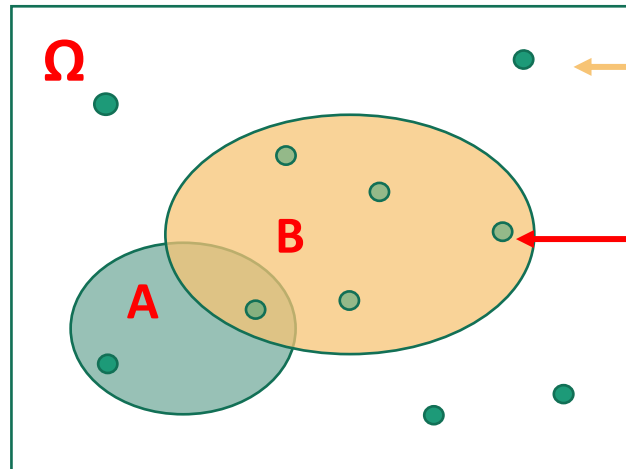
A coin tossing problem

- Given that there were 3 heads in 10 tosses, what is the probability that the first two tosses were heads?
 - event A : the first 2 tosses were heads.
 - event B : 3 out of 10 tosses were heads.
- Second solution: Conditional probability law (on B) is uniform

Assumptions:

- independence
- $P(H) = p$

$$P(k \text{ heads}) = \binom{n}{k} p^k (1-p)^{n-k}$$



← Length 10 sequence

← 3 heads sequences
 $p^3(1-p)^7$

$$\frac{\# \text{ in } (A \cap B)}{\# \text{ in } B} = \frac{8}{\binom{10}{3}}$$

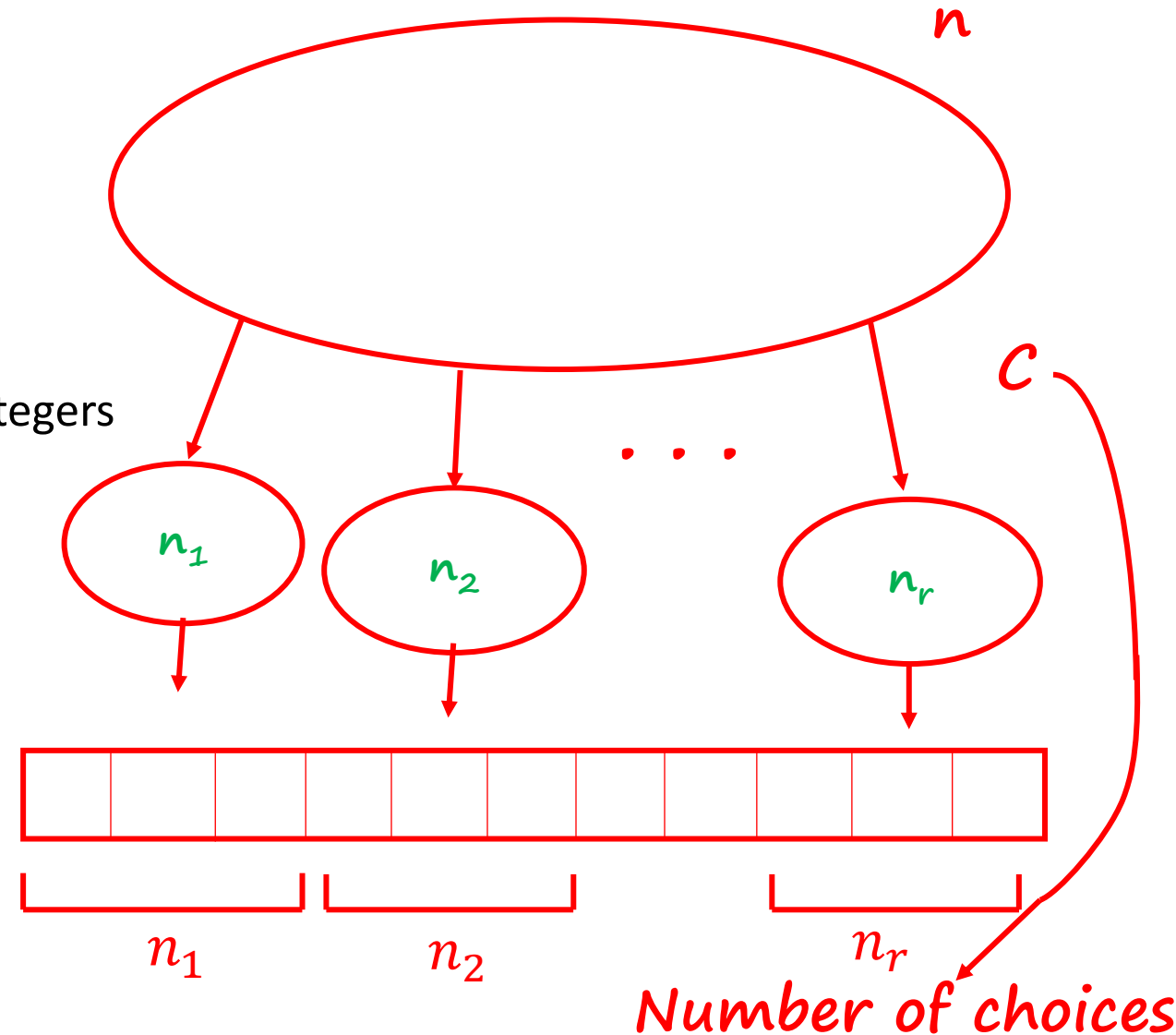
Partitions

- $n \geq 1$ distinct items; $r \geq 1$ persons
Give n_i items to person i
 - Here $n_1, \dots, n_r$ are given nonnegative integers
 - With $n_1 + \dots + n_r = n$ are given nonnegative integers
- Ordering n items: $n!$
 - Deal n_i to each person i , and then order

$$C \cdot n_1! n_2! \dots n_r! = n!$$

Solving for C we get:

$$\text{Number of Choices, } C = \frac{n!}{n_1! n_2! \dots n_r!} \quad (\text{multinomial coefficient})$$



Example: The Birthday Problem

- Given there are n Students in a class, find the P (no 2 birthdays coincide)
 - $1 < n < 365$

Ω = all possible sequences

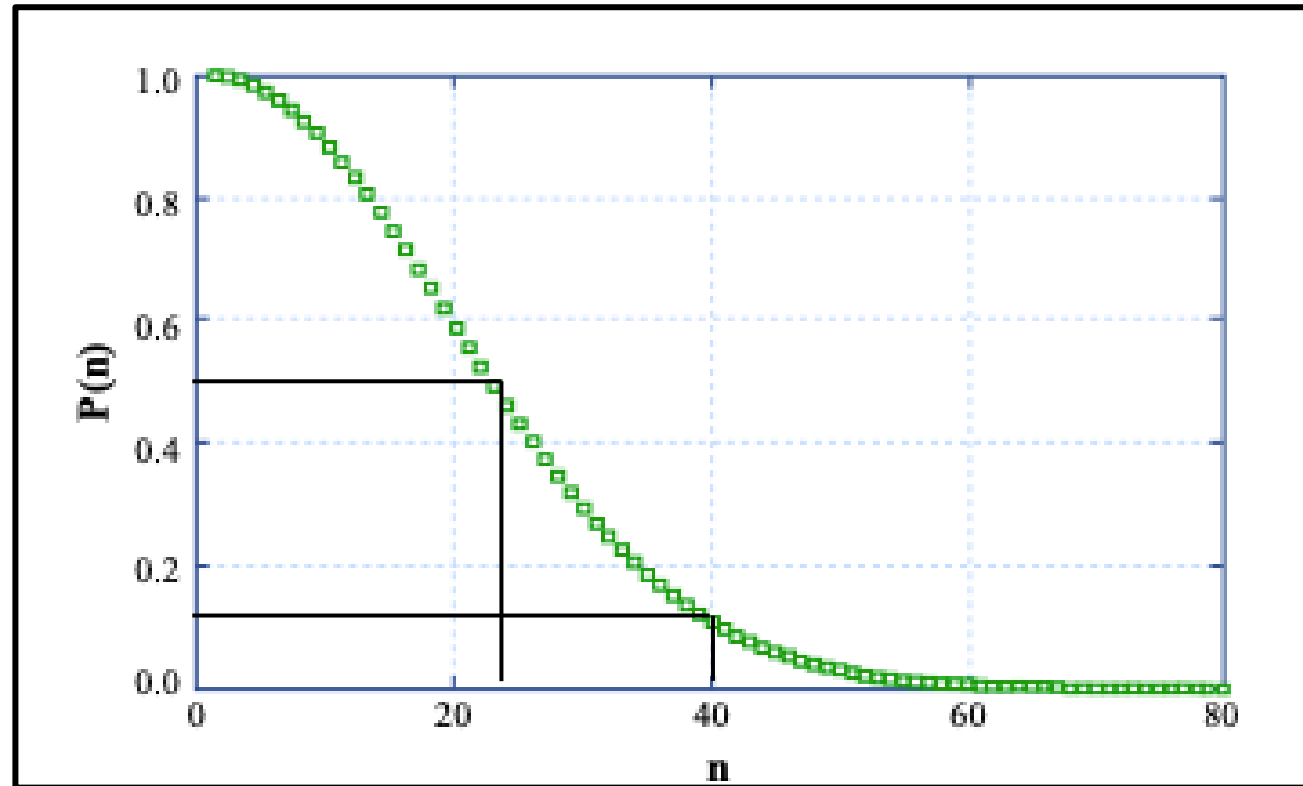
$$|\Omega| = (365)^n$$



$$S = 365 * 364 * \dots * (365 - n + 1) = \frac{365!}{(365 - n)!}$$

$$P(\text{no 2 birthdays coincide}) = \frac{365!}{(365 - n)!} \frac{1}{(365)^n}$$

Example: The Birthday Problem (ctd)



$$P(\text{no 2 birthdays coincide}) = \frac{365!}{(365 - n)!} \frac{1}{(365)^n}$$

Example: 52-card deck, dealt (fairly) to four players.
Find $P(\text{each player gets an ace})$

➤ Total possible Number of outcomes: $\frac{52!}{13! 13! 13! 13!}$

➤ Constructing an outcome with one ace to each person:
(Distribute the aces) $\frac{4!}{1! 1! 1! 1!} = 4 * 3 * 2 * 1$

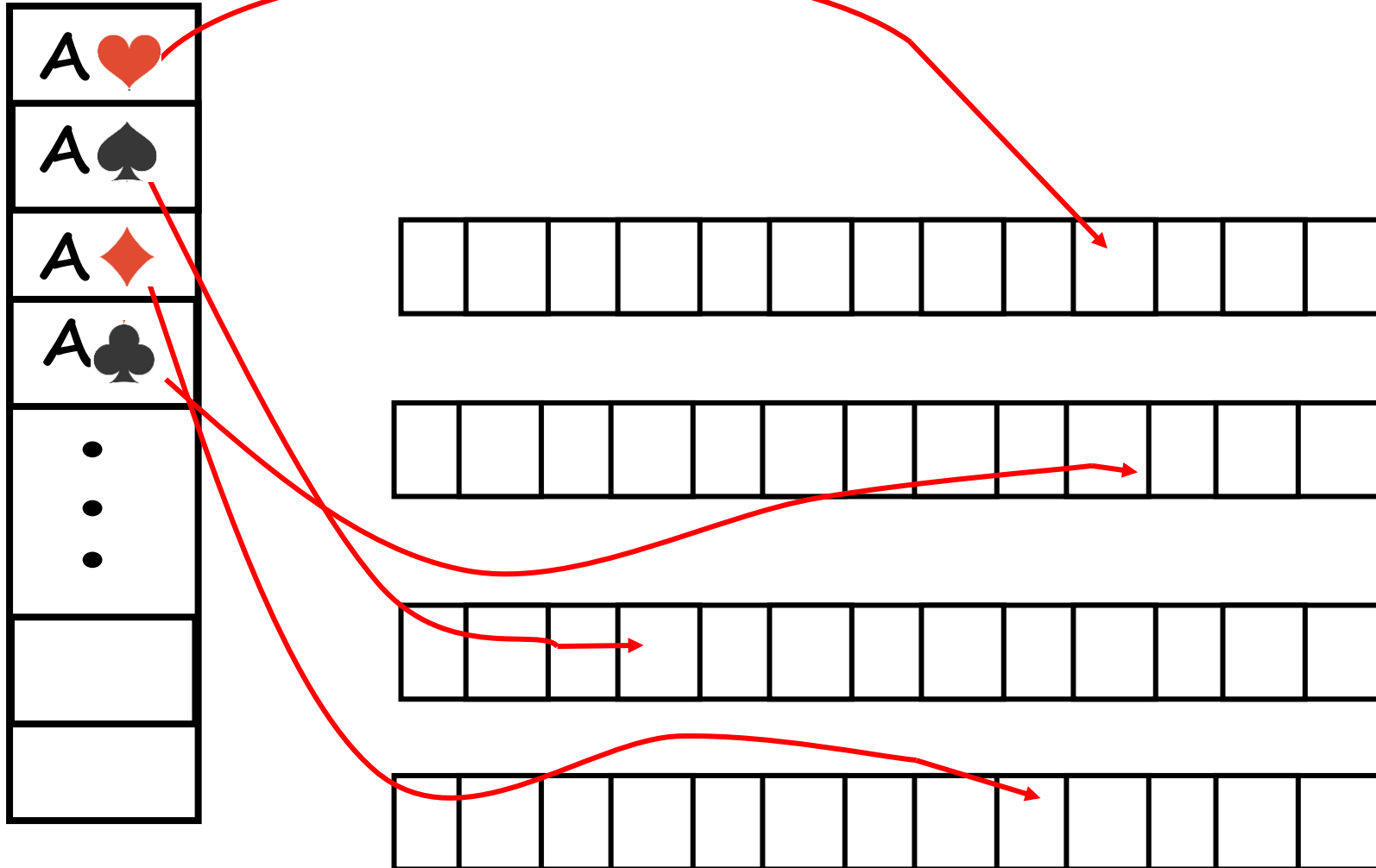
➤ Distribute the remaining 48 cards: $\frac{48!}{12! 12! 12! 12!}$

➤ Answer: $P(\text{each player gets an ace}) = \frac{\frac{4!}{1! 1! 1! 1!} * \frac{48!}{12! 12! 12! 12!}}{\frac{52!}{13! 13! 13! 13!}}$

Example: 52-card deck, dealt (fairly) to four players.
Find $P(\text{each player gets an ace})$

A smart solution

Stack the deck, with 4 aces on top



$$\begin{aligned} & A♥ \quad A♠ \quad A♦ \quad A♣ \\ & \frac{52}{52} * \frac{39}{51} * \frac{26}{50} * \frac{13}{49} \\ & = 0.105 \end{aligned}$$